

2-使用unsloth进行模型CoT微调

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(1) 通过LoRA和4int量化微调DeepSeek

注意到，我们第一部分的工作并没有对模型进行微调，实际上仅依靠模型内部所掌握的知识来作出预测。而希望通过训练的方式让模型避免幻觉，则可以通过将训练集train的样本组成完成的推理链指令，并让模型通过监督微调(supervised fine-tuning, SFT)的办法进行训练。

(2) 降低硬件资源消耗

- 通过LoRA挂载，仅微调LoRA部分
- 使用bitsandbytes进行4int量化模型
- 使用bf16进行训练
- 注意gradient_checkpointing=True, gradient_accumulation=True

(3) Unsloth微调

Unsloth是一个开源框架，旨在使微调大型语言模型（LLM）的速度提高2倍，内存效率更高。

- Unsloth的官网: <https://github.com/unslothai/unsloth>
- 主页上有很多的Demo可以参考
- 可以挂在的model列表: <https://docs.unsloth.ai/get-started/all-our-models>
- 具体安装流程:

```
%%capture
!pip install unsloth
!pip install --force-reinstall --no-cache-dir --no-deps
git+https://github.com/unslothai/unsloth.git
```

因为需要下载Huggingface的大模型，某些大模型如果需要登录的话，可以使用下面的代码

```
from huggingface_hub import login
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()

hf_token = user_secrets.get_secret("HUGGINGFACE_TOKEN")
login(hf_token)
```

(4) 读取模型和tokenizer

接下来，加载DeepSeek-R1-Distill-Llama-8B的Uncloth版本。此外，以4位量化加载模型，以优化内存使用 and 性能。

```
from unsloth import FastLanguageModel

max_seq_length = 2048
dtype = None
load_in_4bit = True

model, tokenizer = FastLanguageModel.from_pretrained(
    model_name = "unsloth/DeepSeek-R1-Distill-Llama-8B",
    max_seq_length = max_seq_length,
    dtype = dtype,
    load_in_4bit = load_in_4bit,
    token = hf_token,
)
```

(5) 直接使用模型推理

参考之前的做法，定义一个提示模板，并包含用于问题和相应生成的占位符，提示将引导模型逐步思考，并提供逻辑清晰、准确的回复。

```
prompt_style = """Below is an instruction that describes a task, paired with
an input that provides further context.

Write a response that appropriately completes the request.

Before answering, think carefully about the question and create a step-by-
step chain of thoughts to ensure a logical and accurate response.
```

```

### Instruction:
You are a medical expert with advanced knowledge in clinical reasoning,
diagnostics, and treatment planning.
Please answer the following medical question.

### Question:
{}

### Response:
<think>{}"""

```

为了演示，使用一个医学的问题，并使用上述模版进行拼接，提交给模型直接进行推理。

```

question = "A 61-year-old woman with a long history of involuntary urine loss
during activities like coughing or sneezing but no leakage at night undergoes
a gynecological exam and Q-tip test. Based on these findings, what would
cystometry most likely reveal about her residual volume and detrusor
contractions?"

FastLanguageModel.for_inference(model)
inputs = tokenizer([prompt_style.format(question, "")],
return_tensors="pt").to("cuda")

outputs = model.generate(
    input_ids=inputs.input_ids,
    attention_mask=inputs.attention_mask,
    max_new_tokens=1200,
    use_cache=True,
)

response = tokenizer.batch_decode(outputs)
print(response[0].split("### Response:")[1])

```

即使没有微调，DeepSeek依然可以顺利地进行思维链，并产生推理过程，提交最终答案。注意到，输出结果中<think></think>之间所包裹的内容就是推理的内容。

那为什么我们还需要进行微调呢？仔细观察，推理过程虽然详细，但冗长而不简洁。此外，最终答案以要点格式呈现，这与要微调的数据集的结构和风格有所不同。

```

<think>
Okay, so I have this medical question to answer. Let me try to break it down.
The patient is a 61-year-old woman with a history of involuntary urine loss
during activities like coughing or sneezing, but she doesn't leak at night.
She's had a gynecological exam and a Q-tip test. I need to figure out what

```

cystometry would show regarding her residual volume and detrusor contractions.

First, I should recall what I know about urinary incontinence. Involuntary urine loss during activities like coughing or sneezing makes me think of stress urinary incontinence. Stress incontinence typically happens when the urethral sphincter isn't strong enough to resist increased abdominal pressure from activities like coughing, laughing, or sneezing. This usually affects women, especially after childbirth when the pelvic muscles and ligaments are weakened.

The Q-tip test is a common diagnostic tool for stress urinary incontinence. The test involves inserting a Q-tip catheter, which is a small balloon catheter, into the urethra. The catheter is connected to a pressure gauge. The patient is asked to cough, and the pressure reading is taken. If the pressure is above normal (like above 100 mmHg), it suggests that the urethral sphincter isn't closing properly, which is a sign of stress incontinence.

So, based on the history and the Q-tip test, the diagnosis is likely stress urinary incontinence. Now, moving on to what cystometry would show. Cystometry, also known as a filling cystometry, is a diagnostic procedure where a catheter is inserted into the bladder, and the bladder is filled with a liquid to measure how much it can hold (residual volume) and how it responds to being filled (like during a cough or sneeze). This helps in assessing the capacity and compliance of the bladder.

In a patient with stress incontinence, the bladder's capacity might be normal, but the sphincter's function is impaired. So, during the cystometry, the residual volume might be within normal limits because the bladder isn't overfilled. However, when the patient is asked to cough or perform a Valsalva maneuver, the detrusor muscle (the smooth muscle layer of the bladder) might not contract effectively, leading to an increase in intra-abdominal pressure, which might cause leakage.

Wait, but detrusor contractions are usually associated with voiding. In stress incontinence, the issue isn't with the detrusor contractions but with the sphincter's inability to prevent leakage. So, during cystometry, the detrusor contractions would be normal because they are part of the normal voiding process. However, the problem is that the sphincter doesn't close properly, leading to leakage.

So, putting it all together, the residual volume might be normal, but the detrusor contractions would be normal as well. The key finding would be the impaired sphincter function leading to incontinence, which is typically

demonstrated during the Q-tip test and clinical history. Therefore, the cystometry would likely show normal residual volume and normal detrusor contractions, but the underlying issue is the sphincter's inability to prevent leakage.

</think>

Based on the provided information, the cystometry findings in this 61-year-old woman with stress urinary incontinence would likely demonstrate the following:

1. **Residual Volume**: The residual volume would be within normal limits. This is because the bladder's capacity is typically normal in cases of stress incontinence, where the primary issue lies with the sphincter function rather than the bladder's capacity.
2. **Detrusor Contractions**: The detrusor contractions would also be normal. These contractions are part of the normal voiding process and are not impaired in stress urinary incontinence. The issue is not with the detrusor muscle but with the sphincter's inability to prevent leakage.

In summary, the key findings of the cystometry would be normal residual volume and normal detrusor contractions, highlighting the sphincteric defect as the underlying cause of the incontinence.<|end__of__sentence|>

(6) 读取并处理数据

微调与推理不同，推理进需要写到### Response: 位置即可。微调则需要组成全部的推理指令，并通过SFT让模型进行下词预测的推理训练即可。因此，稍微调整一下指令模版：

```
train_prompt_style = """Below is an instruction that describes a task, paired
with an input that provides further context.

Write a response that appropriately completes the request.

Before answering, think carefully about the question and create a step-by-
step chain of thoughts to ensure a logical and accurate response.

### Instruction:

You are a medical expert with advanced knowledge in clinical reasoning,
diagnostics, and treatment planning.

Please answer the following medical question.

### Question:

{}
```

```

### Response:
<think>
{}
</think>
{} """

```

编写Python函数，该函数将在数据集中创建一个text列，该列由训练提示样式组成。用问题、思维链和答案填充占位符。

```

EOS_TOKEN = tokenizer.eos_token # Must add EOS_TOKEN

def formatting_prompts_func(examples):
    inputs = examples["Question"]
    cots = examples["Complex_CoT"]
    outputs = examples["Response"]
    texts = []
    for input, cot, output in zip(inputs, cots, outputs):
        text = train_prompt_style.format(input, cot, output) + EOS_TOKEN
        texts.append(text)
    return {
        "text": texts,
    }

```

为了演示，从FreedomIntelligence/medical-o1-reasoning-SFT数据集中加载前500个样本，该数据集可在Hugging Face中心获得。之后，将使用formatting_prompts_func函数映射文本列。

针对具体任务，例如上次使用的SST进行数据获取，可以通过大模型完成。例如：使用GPT-4o的API，通过提示让其产生上述的思维链过程，并通过文本处理拼接后，给小模型来进行指令微调时使用。

```

from datasets import load_dataset
dataset = load_dataset("FreedomIntelligence/medical-o1-reasoning-SFT", "en",
split = "train[0:500]", trust_remote_code=True)
dataset = dataset.map(formatting_prompts_func, batched = True,)
dataset["text"][0]

```

拼接后的提示指令为：

```

"Below is an instruction that describes a task, paired with an input that
provides further context. \nWrite a response that appropriately completes the
request. \nBefore answering, think carefully about the question and create a
step-by-step chain of thoughts to ensure a logical and accurate

```

response.\n\n### Instruction:\nYou are a medical expert with advanced knowledge in clinical reasoning, diagnostics, and treatment planning.

\nPlease answer the following medical question. \n\n### Question:\nA 61-year-old woman with a long history of involuntary urine loss during activities like coughing or sneezing but no leakage at night undergoes a gynecological exam and Q-tip test. Based on these findings, what would cystometry most likely reveal about her residual volume and detrusor contractions?\n\n### Response:\n<think>\nOkay, let's think about this step by step. There's a 61-year-old woman here who's been dealing with involuntary urine leakages whenever she's doing something that ups her abdominal pressure like coughing or sneezing. This sounds a lot like stress urinary incontinence to me. Now, it's interesting that she doesn't have any issues at night; she isn't experiencing leakage while sleeping. This likely means her bladder's ability to hold urine is fine when she isn't under physical stress. Hmm, that's a clue that we're dealing with something related to pressure rather than a bladder muscle problem. \n\nThe fact that she underwent a Q-tip test is intriguing too. This test is usually done to assess urethral mobility. In stress incontinence, a Q-tip might move significantly, showing urethral hypermobility. This kind of movement often means there's a weakness in the support structures that should help keep the urethra closed during increases in abdominal pressure. So, that's aligning well with stress incontinence.\n\nNow, let's think about what would happen during cystometry. Since stress incontinence isn't usually about sudden bladder contractions, I wouldn't expect to see involuntary detrusor contractions during this test. Her bladder isn't spasming or anything; it's more about the support structure failing under stress. Plus, she likely empties her bladder completely because stress incontinence doesn't typically involve incomplete emptying. So, her residual volume should be pretty normal. \n\nAll in all, it seems like if they do a cystometry on her, it will likely show a normal residual volume and no involuntary contractions. Yup, I think that makes sense given her symptoms and the typical presentations of stress urinary incontinence.\n</think>\nCystometry in this case of stress urinary incontinence would most likely reveal a normal post-void residual volume, as stress incontinence typically does not involve issues with bladder emptying. Additionally, since stress urinary incontinence is primarily related to physical exertion and not an overactive bladder, you would not expect to see any involuntary detrusor contractions during the test.<|end__of__sentence|>"

(7) 获取并设置模型

使用get_peft_model来挂载Lora

```

model = FastLanguageModel.get_peft_model(
    model,
    r=16,
    target_modules=[
        "q_proj",
        "k_proj",
        "v_proj",
        "o_proj",
        "gate_proj",
        "up_proj",
        "down_proj",
    ],
    lora_alpha=16,
    lora_dropout=0,
    bias="none",
    use_gradient_checkpointing="unsloth", # True or "unsloth" for very long context
    random_state=3407,
    use_rslora=False,
    loftq_config=None,
)

```

这里特别注意`use_gradient_checkpointing`，在具体运算是进行activation的存储，这样可以进一步降低显存消耗，但会损耗一些计算资源。

接下来，通过提供模型、标记器、数据集和其他重要的训练参数来设置训练参数和训练器，这些参数将优化我们的微调过程。

```

from trl import SFTTrainer
from transformers import TrainingArguments
from unsloth import is_bfloat16_supported

trainer = SFTTrainer(
    model=model,
    tokenizer=tokenizer,
    train_dataset=dataset,
    dataset_text_field="text",
    max_seq_length=max_seq_length,
    dataset_num_proc=2,
    args=TrainingArguments(
        per_device_train_batch_size=2,
        gradient_accumulation_steps=4,
        # Use num_train_epochs = 1, warmup_ratio for full training runs!
        warmup_steps=5,
    )
)

```



```

max_steps=60,
learning_rate=2e-4,
fp16=not is_bfloat16_supported(),
bf16=is_bfloat16_supported(),
logging_steps=10,
optim="adamw_8bit",
weight_decay=0.01,
lr_scheduler_type="linear",
seed=3407,
output_dir="outputs",
),
)

```

开始训练：

```
trainer_stats = trainer.train()
```

 [60/60 44:06, Epoch 0/1]

Step	Training Loss
10	1.919000
20	1.461800
30	1.402500
40	1.309000
50	1.344400
60	1.314100

(8) 微调后的模型推理

为了比较结果，向微调模型提出与以前相同的问题，以查看发生了什么变化。

```
question = "A 61-year-old woman with a long history of involuntary urine loss
during activities like coughing or sneezing but no leakage at night undergoes
a gynecological exam and Q-tip test. Based on these findings, what would
cystometry most likely reveal about her residual volume and detrusor
contractions?"
```

```
FastLanguageModel.for_inference(model) # Unsloth has 2x faster inference!
inputs = tokenizer([prompt_style.format(question, "")],
return_tensors="pt").to("cuda")
```

```
outputs = model.generate(
```

```
input_ids=inputs.input_ids,
attention_mask=inputs.attention_mask,
max_new_tokens=1200,
use_cache=True,
)
response = tokenizer.batch_decode(outputs)
print(response[0].split("### Response:")[1])
```

这样回复要好得多，也更加准确。思路很直接，答案很简单。如此，微调起到了一定的作用。

<think>

Okay, so let's think about this. We have a 61-year-old woman who's been dealing with involuntary urine loss during things like coughing or sneezing, but she's not leaking at night. That suggests she might have some kind of problem with her pelvic floor muscles or maybe her bladder.

Now, she's got a gynecological exam and a Q-tip test. Let's break that down. The Q-tip test is usually used to check for urethral obstruction. If it's positive, that means there's something blocking the urethra, like a urethral stricture or something else.

Given that she's had a positive Q-tip test, it's likely there's a urethral obstruction. That would mean her urethra is narrow, maybe due to a stricture or some kind of narrowing. So, her bladder can't empty properly during activities like coughing because the urethral obstruction is making it hard.

Now, let's think about what happens when her bladder can't empty. If there's a urethral obstruction, the bladder is forced to hold more urine, increasing the residual volume. That's because her bladder doesn't empty completely. So, her residual volume is probably increased.

Also, if her bladder can't empty properly, she might have increased detrusor contractions. These contractions are usually stronger to push the urine out. So, we expect her detrusor contractions to be increased.

Putting it all together, if she has a urethral obstruction and a positive Q-tip test, we'd expect her cystometry results to show increased residual volume and increased detrusor contractions. That makes sense because of the obstruction and how her bladder is trying to compensate by contracting more.

</think>

Based on the findings of the gynecological exam and the positive Q-tip test, it is most likely that the cystometry would reveal increased residual volume and increased detrusor contractions. The positive Q-tip test indicates

urethral obstruction, which would force the bladder to retain more urine, thereby increasing the residual volume. Additionally, the obstruction can lead to increased detrusor contractions as the bladder tries to compensate by contracting more to expel the urine.< | end__of__sentence | >

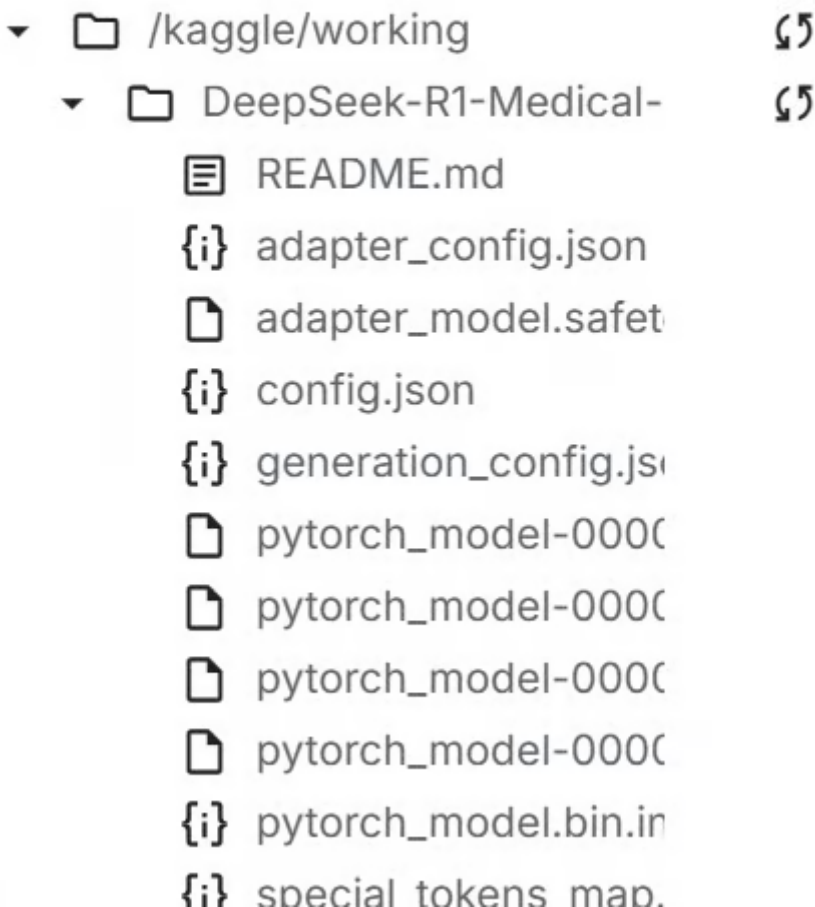
(9) 保存模型，并用LM Studio进行挂载

保存模型，需要保存LoRA适配器、全量模型和Tokenizer：

```
new_model_local = "DeepSeek-R1-Medical-COT"
model.save_pretrained(new_model_local)
tokenizer.save_pretrained(new_model_local)

model.save_pretrained_merged(new_model_local, tokenizer, save_method =
"merged_16bit",)
```

Output (15.4GiB / 19.5GiB)



另一种办法可以把结果传到服务器上，放到Huggingface Hub中

```
new_model_online = "<username>/DeepSeek-R1-Medical-COT"  
model.push_to_hub(new_model_online)  
tokenizer.push_to_hub(new_model_online)  
  
model.push_to_hub_merged(new_model_online, tokenizer, save_method =  
    "merged_16bit")
```

注意把上面的<username>部分换成是你的用户名。

(10) 记录结果

试着用更大的模型合成SST的数据，引导小模型版本的Llama、Qwen、Gemma、Phi-4和Mistral进行微调，并统计下SST-2指令学习的准确度。再和之前的实验结果对比一下。